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DEVELOPMENT AND EVALUATION OF SUSTAINED RELEASE MATRIX TABLETS OF NICLOSAMIDE FOR ANTHELMINTIC THERAPY

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ABSTRACT:

Helminthiasis is a common human health disease, which is mostly managed using locally acting antihelmintic drugs such as Niclosamide. The downside with conventional immediate-release Niclosamide tablets is, however, that they are prone to rapid dose dumping.This brings about intensive gastro intestinal irritation and necessitates multiple dosing every few hours, which eventually leads to low compliance by the patient. The present research is intended to develop and test SR matrix tablets of Niclosamide at once daily to increase the local contact time of this medication in the gastrointestinal tract, decrease any side effects, and enhance the overall therapeutic effectiveness.The wet granulation method was used to prepare the matrix tablets at the following concentrations of the release-retarding polymers; Hydroxypropyl Methylcellulose (HPMC K100M), Guar Gum, and Ethyl Cellulose.FTIR pre-formulation studies ensured that the drug was in good chemical interaction with the excipients.All the prepared batches (F1-F8) were tested in terms of pre-compression flow characteristics and post-compression quality control parameters, which were observed to be within the usual pharmacopoeial limits quite satisfactorily.Dissolution experiments in vitro over 12 hours showed that the optimized batch of the formulation F2, which included 150 mg of HPMC K100M, was able to provide a highly controlled and almost complete drug release of 97.4.Kinetic behavior of F2-- Experiments that were made kinetically showed the drug release was most effectively described by the Higuchi model with an anomalous (non-Fickian) transport mechanism by diffusion as well as polymer erosion.Also, in vitro anthelmintic bioassays on adult Indian earthworms (Pheretima posthuma) revealed that the optimized formulation did not lose any biological activity, and was able to induce paralysis and death in the parasites. A three-month stability experiment indicated that there were no significant changes in physical and chemical properties. The sustained-release matrix tablets made using HPMC are therefore a very convenient and easy to use alternative to the current conventional therapy that is effective in removing tapeworms in the intestine.

Keywords: *Niclosamide, Sustained release, Matrix tablets, Helminthiasis, Anthelmintic activity, HPMC K100M, Wet granulation.*

Introduction

One of the most common public health problems in the modern world is the parasitic worm infection which is also known as Helminthiasis. These infections are especially pronounced in the third world countries such as India where clean drinking water, poor sanitation and human populations facilitate easy transmission of the parasites (Hotez et al., 2008). The diseases cause malnutrition, acute anemia, slower physical and mental growth in children and the overall loss of working power of the adult population per day (Jourdan et al., 2018). In the rural regions, the disease contributes immensely to low agricultural output and children miss most of their school days. Although the school level is where most mass deworming programs are implemented by the government, the intestinal worms still lead to high sickness rates. Thus, it is highly essential that superior and patient friendly therapies should be discovered to totally rid the human body of these parasites (Sharma & Kumar, 2020).

In order to cure these parasitic infections, physicians use a category of medication known as anthelmintics. Niclosamide is a famous, old drug which is widely utilized in the range of treatment choices. It has been used, at least since the 1950s, to treat intestinal infections caused by a range of different varieties of tapeworms, including Taenia saginata (beef tapeworm) Taenia solium (pork tapeworm), Hymenolepi nana (dwarf tapeworm) and Diphyllobothrium latum (fish tapeworm) (Pearson & Hewlett, 1985).The medication destroys the worms in that they lose their ability to generate own energy. In particular, the oxidative phosphorylation of the mitochondria of the tapeworm is uncoupled by Niclosamide. This will mean that the worm would not be able to produce Adenosine Triphosphate (ATP) (Andrews et al.1983). In case this life force is absent, the worm dies in the intestine and then safely comes out of the patient in the form of stool. Other medical uses Niclosamide is also currently under investigation in other uses like an antibacterial agent, antiviral, but its primary clinical use is in intestinal parasites.

The use of the standard conventional tablets of Niclosamide does have some serious minuses to practical use though. First of all, Niclosamide is a drug that is highly insoluble in water chemically (Lodagekar et al., 2019). Because it dissolves in the liquids of the stomach not readily, it acts topographically in the intestines. Whereas in reality the presence of this local effect is required to treat intestinal worms (because the medication has to stay in the gastrointestinal tract to contact the worms) the conventional fast-dissolving pills discharged the medication at a remarkable rate. In the event that sudden loading of the drug in the stomach takes place, it may be very irritating thus causing pain, nausea, and vomiting to the patient (Singh & Kaur, 2021).

The normal medicine is absorbed too fast through bowels and hence the worms are only exposed to the medicine at a very low time. Patients have to take repeated large doses in a few days in order to clear all the worms. Large and frequent dosage leads to medication non-adherence and this is why there are a number of individuals that simply abandon intake of the drug before they have fully recovered (Patel & Desai, 2017).

In order to give solutions to these clinical problems in practice, pharmaceutical researchers are laying much emphasis on SR delivery systems. The concept of sustained-release system is to deliver the active drug in a slow and regular manner Over a prolonged time, usually over 12- 24 hours (Chien YW 1992). A sustained-release product is very advantageous in the case of a locally acting anthelmintic drug such as Niclosamide. The slow rate of release of the drug into the stomach and intestines ensures that the local level of the medicine remains so elevated that it paralyzes and kills the parasites in a few consecutive hours (Kumar & Prajapati, 2020). It is deadly to the worms and the clinical effect of the drug is significantly increased because this time of contact is extended. Meanwhile, slow release ensures that the body is introduced to the drug slowly and does away with the high concentration of drugs that causes stomach upset thus making the entire course of treatment far safer and more tolerable (Gupta & Sharma , 2018).

What is even better is the fact that sustained-release tablet means that the patient will only need to administer his medication to himself once a day, which makes it much more probable that the patient will be able to stick with his treatment regimen (Jain, 2001). Among the other approaches to creating a sustained-release medicine, matrix tablets are one of the most common and successful approaches to the pharmaceutical industry so far. The active drug powder is then mixed with a polymer a substance that acts as a holding mesh in equal proportions and then pressed into a solid pill to make a matrix pill (Siepmann & Peppas, 2001). Once the patient is taking this pill, the polymer slowly swells once exposed to stomach fluids and a layer of thick gel is formed. Slowly, a drug then passes through this gel layer during several hours (Reddy et al., 2003) in many hours. Matrix tablets are also quite desirable by the drug manufacturers, as they can be produced with the aid of the traditional equipment, are highly cost-effective and predictable in releasing the drugs.

Designing, fabricating and testing sustained-release matrix tablets of Niclosamide using a mixture of hydrophilic and hydrophobic polymers is the main objective of the research. In particular, the study will be aimed at the optimization of the formulation that can be used to promote long-term gastrointestinal absorption of the medication. Also, the paper will assess the physicochemical properties of the formulation, it's in vitro release kinetics, and explore its local anthelmintic activity and stability at long-term. The benefit of this method

is to increase treatment efficacy against intestinal tapeworms and increase patient compliance since there is a simplified dosing schedule of once a day only.

Materials and Methods

Materials:

Niclosamide was obtaining from Yarrow Chem Products, Mumbai, India. Polymers that release gradually (HPMC K100M, Guar Gum, Ethyl Cellulose), and formulation excipients (MCC PH 102, PVP K-30, Magnesium Stearate, Talc) were obtained in LobaChemie Pvt. Ltd., Mumbai. Analytical grade chemical and solvents were used during the project work.

Pre-formulation and Analytical Method:

Standard procedures were used to determine the physicochemical characteristics of Niclosamide such as organoleptic properties, solubility and melting point (Al-Badr & Hafez, 2018). Fourier Transform Infrared Spectroscopy (FTIR) was used to evaluate the drug-polymer compatibility in a scanning range of 4000 to 400 cm^{-1}(Mendham et al., 2000). To quantify the drug, a UV-Visible spectrophotometric technique was designed and calibrated at 333 nm (Skoog et al., 2017)

Formulation of Matrix Tablets:

Eight batches (F1-F8) of sustained-release matrix tablets, in which 500 mg niclosamide was used, were prepared using the wet granulation method. The formulation matrices were different depending on the type and concentration of the rate-retarding polymer (HPMC, Guar Gum, or Ethyl Cellulose). The powdered mixes were granulated using a PVP K-30 solution in isopropyl alcohol. The resultant granules were dried at 50o C, sieved through a 20 mesh screen, lubricated with talc and magnesium stearate, and pressed in a rotary tablet press. (Aulton & Taylor, 2017)

Table 1 Composition of Niclosamide Sustained Release Matrix Tablets (F1 - F8)

Ingredients (mg/tab)	Function	F1	F2	F3	F4	F5	F6	F7	F8

Niclosamide	Active Pharmaceutical Ingredient	500	500	500	500	500	500	500	500
HPMC K100M	Hydrophilic Matrix Polymer	100	150	200	--	--	--	--	--
Guar Gum	Natural Matrix Polymer	--	--	--	100	150	200	--	--
Ethyl Cellulose	Hydrophobic Polymer	--	--	--	--	--	--	100	150
MCC (Avicel pH 102)	Diluent / Binder	190	140	90	190	140	90	190	140
Magnesium Stearate	Lubricant	5	5	5	5	5	5	5	5
Talc	Glidant	5	5	5	5	5	5	5	5
Total Weight (mg)	--	**800**	**800**	**800**	**800**	**800**	**800**	**800**	**800**

Pre- and Post-Compression Evaluation:

The lubricated granules were measured on micromeritic properties (angle of repose, bulk/tapped density, Carr Compressibility Index, and Hausner ratio) to guarantee sufficient flowability before compression (United States Pharmacopeial Convention, 2023).After compression, the tablets were put through standard tests of pharmacopoeial quality control to measure variation in weight, thickness, hardness, and friability (limit less than 1%), and uniformity in overall drug content (Indian Pharmacopoeia Commission,2018).

In Vitro Dissolution and Release Kinetics:

The dissolution profiling was done on a USP Dissolution Apparatus II (Paddle Method) at 37 ± 0.5°C during the first 2 hours of the study, then changing to Simulated Intestinal Fluid (pH 6.8/7.4) during the rest of the 24 hours(Costa & Sousa Lobo,2001).Aliquots were analyzed periodically spectrophotometrically.In order to

explain the mechanism of drug release, empirical data on dissolution was mathematically modelled using Zero-order, First-order, Higuchi and Korsmeyer-Peppas kinetic models (Dash et al., 2010)

Anthelmintic Efficacy and Stability Studies

The in vitro anthelmintic activity of the optimized formulation was assessed using adult Indian earthworms (Pheretima posthuma). Efficacy was quantified by recording the mean time to paralysis and mean time to death, benchmarked against a standard reference solution (Vidyarthi et al., 2012) Finally, accelerated stability testing of the optimized batch was conducted in accordance with ICH Q1A (R2) guidelines (40°C ± 2°C / 75% ± 5% RH) over a 3-month period to verify physicochemical integrity (International Council for Harmonization, 2003).

Results and Discussion

Pre-formulations Studies

Organoleptic Properties and Melting Point:

Evaluation of the sample of Niclosamide for its physical properties indicated that it was an odorless, pale yellow to pale-yellow color crystalline powder. The melting point was determined using the capillary method and the determined melting point was between 226°C and 229°C, within the pharmacopoeial-defined melting point of 225°C – 230°C which supports the conclusion that the sample of Niclosamide is manufactured with good purity.

Solubility Profile

Evaluation of the solubility profile was conducted on the compound Niclosamide. Based on the analysis, the drug is classified as BCS Class II based on no solubility (>0.1N HCl and in alkaline medium) based on no solubility in the media tested. However, when tested in a phosphate buffer (7.4 pH), it was determined that Niclosamide had some solubility. This characteristic necessitates an effective formulation strategy to maintain consistent release rates, even with limited solubility.

Drug-Excipient Compatibility (FTIR Analysis)

The Fourier Transform Infrared (FTIR) spectra were obtained for the pure compound Niclosamide. The FTIR spectrum contained several key peaks that identify the compound, the peaks were located at 3578 cm^{-1} (O-H stretch), at 3300 cm^{-1} (N-H stretch), and at 1650 cm^{-1} (C=O stretch of amide). The FTIR spectra of the physical combination of Niclosamide with HPMC K100M, Guar Gum, and Ethyl Cellulose retained all of the essential key peaks of the pure compound; no peaks shifted out in a frequency or became absent, suggesting no chemical interaction has occurred between the selected polymers and Niclosamide, thus allowing the use of these polymers in formulation development

Evaluation of Granules (Pre-Compression Parameters)

The flow properties of the granules prepared by wet granulation were evaluated to ensure uniform die filling during compression. The results for all 8 batches are summarized in Table 2.

Table 2 Micromeritic Properties of Granules

Formulation Code	Angle of Repose (°)	Bulk Density (g/mL)	Tapped Density (g/mL)	Carr's Index (%)	Hausner's Ratio
F1	26.4 ± 1.2	0.42 ± 0.02	0.48 ± 0.01	12.50	1.14
F2	27.1 ± 0.8	0.44 ± 0.01	0.51 ± 0.03	13.72	1.15
F3	28.5 ± 1.5	0.43 ± 0.02	0.52 ± 0.02	17.30	1.20
F4	27.8 ± 1.1	0.45 ± 0.01	0.53 ± 0.02	15.09	1.17
F5	28.1 ± 1.0	0.44 ± 0.02	0.51 ± 0.01	13.72	1.15
F6	29.0 ± 1.2	0.42 ± 0.01	0.50 ± 0.02	16.00	1.19
F7	26.8 ± 0.9	0.46 ± 0.02	0.52 ± 0.01	11.53	1.13

F8	27.5 ± 1.3	0.45 ± 0.01	0.52 ± 0.02	13.46	1.15

Discussion: The Range of Repose for all forms of Niclosamide was between 26.4° and 29.0° which is below the 30° mark we see as the limit which in turn reports very good to excellent flow properties. Also we saw Carr's Index to be under 18% and Hausner's Ratio below 1.25 for all batches. This indicates that the wet granulation process did in fact improve the flow ability of Niclosamide which is a very cohesive material and also that we had minimal weight variation during tablet compression which was true for all types of polymers used.

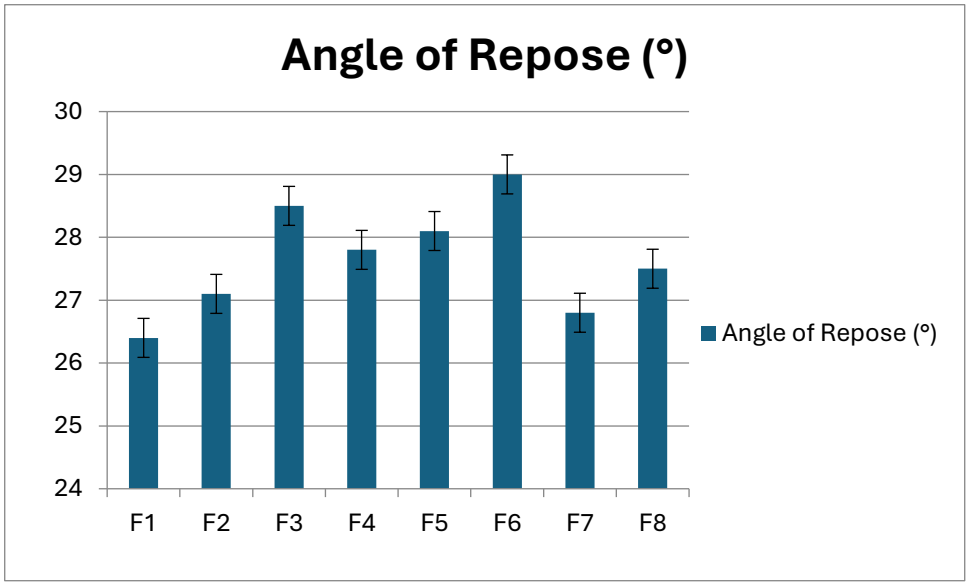


Fig.1: Angle of Repose of Granules for Different Formulations

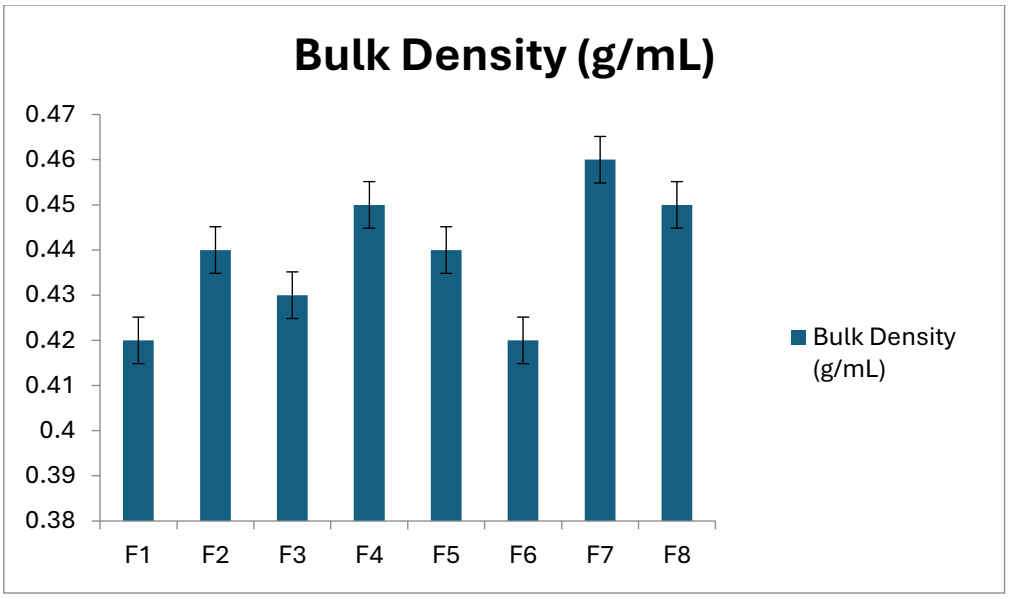


Fig. 2: Bulk Density of Granules For Different Formulations

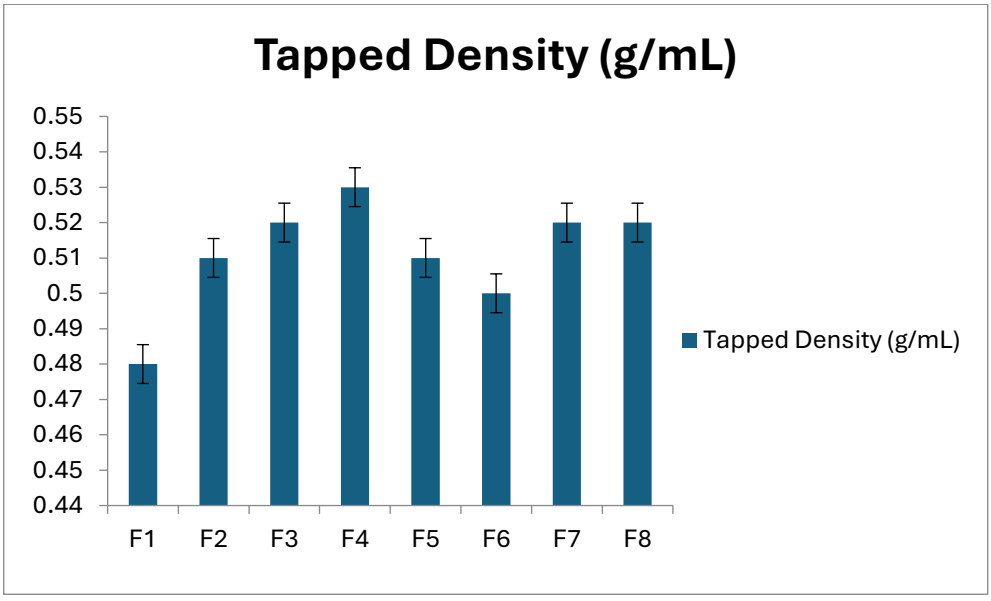


Fig.3: Tapped Density of granules for Different Formulations

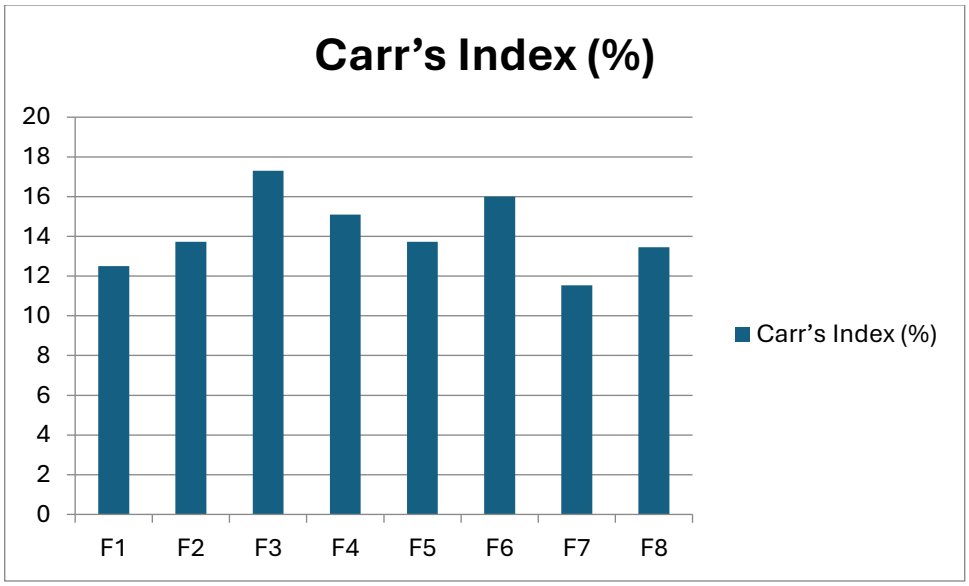


Fig.4: Showing Carr's Index of Granules for Different Formulations

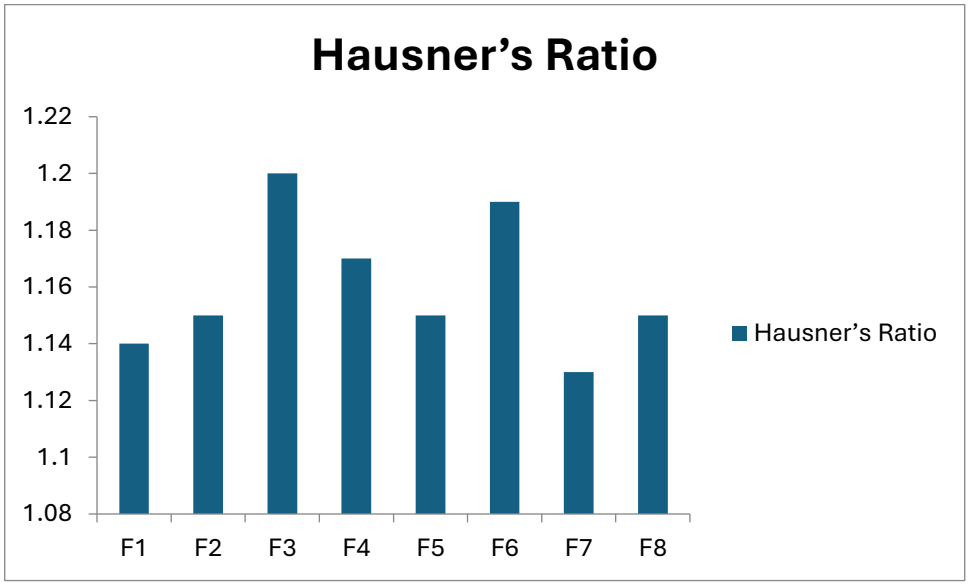


Fig.5: Showing Hausner'sRatio of Granules for Different Formulation

Evaluation of Matrix Tablets (Post-Compression Parameters)

The compressed tablets were evaluated for physicochemical properties. The data is presented in Table 3.

Table 3 Physicochemical Evaluation of Matrix Tablets

Formulation Code	Weight Variation (mg) (n=20)	Hardness (kg/cm²) (n=6)	Thickness (mm) (n=10)	Friability (%) (n=10)	Drug Content (%) (n=10)
F1	798 ± 4.2	5.8 ± 0.4	4.5 ± 0.1	0.65	98.4 ± 1.2
F2	801 ± 3.5	6.2 ± 0.3	4.6 ± 0.2	0.52	99.1 ± 0.8
F3	802 ± 5.1	6.5 ± 0.5	4.7 ± 0.1	0.48	98.9 ± 1.1
F4	799 ± 3.8	6.1 ± 0.4	4.5 ± 0.2	0.58	99.5 ± 0.9
F5	801 ± 4.0	6.0 ± 0.3	4.6 ± 0.1	0.55	98.8 ± 1.0
F6	803 ± 4.5	6.4 ± 0.4	4.7 ± 0.2	0.49	99.2 ± 1.2
F7	798 ± 3.5	5.9 ± 0.3	4.5 ± 0.1	0.60	98.5 ± 0.8
F8	800 ± 4.2	6.3 ± 0.5	4.6 ± 0.1	0.51	99.0 ± 1.1

Discussion:

- All weight variation tests complied with pharmacopoeial standards, with all batches remaining well within ±5% deviation.
- The hardness of all batches maintained between 5.8 and 6.5 kg/cm² providing adequate mechanical strength. All batches showed a friability of less than 1% indicating that the tablets would withstand mechanical shock during packaging and transport.

- Drug content assay showed uniform distribution of Niclosamide between 98.4% and 99.5%. This supports the fact that geometric mixing was performed efficiently when using the different polymers and granulation methods

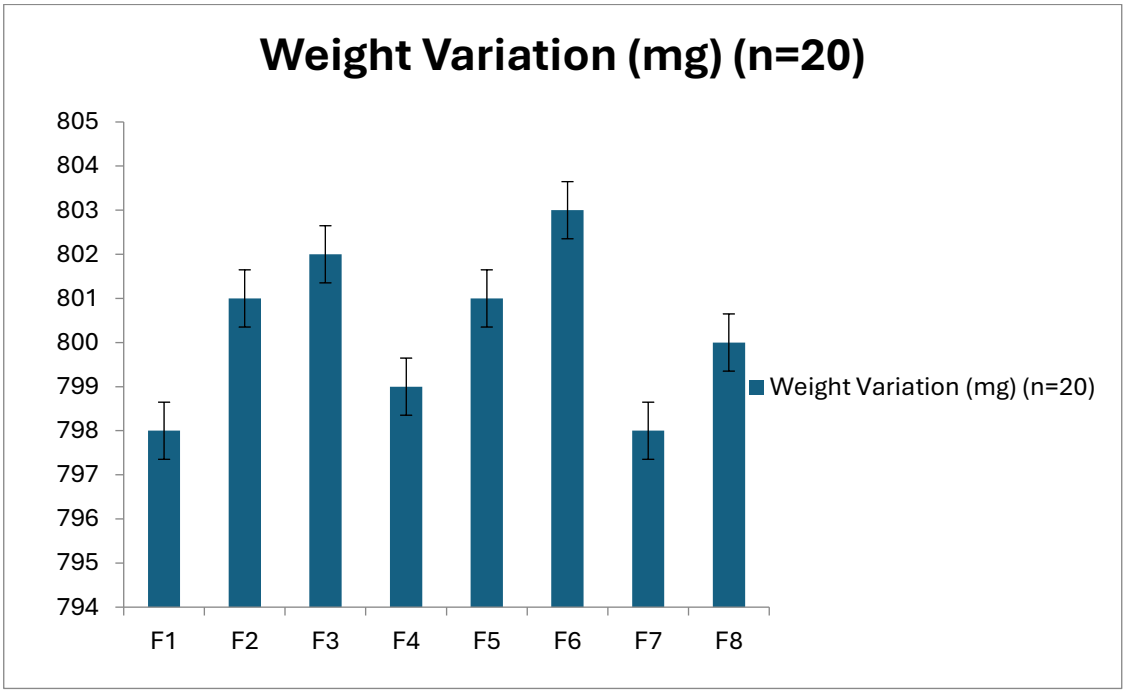


Fig.6: Showing weight variation of matrix tablet for different formulations

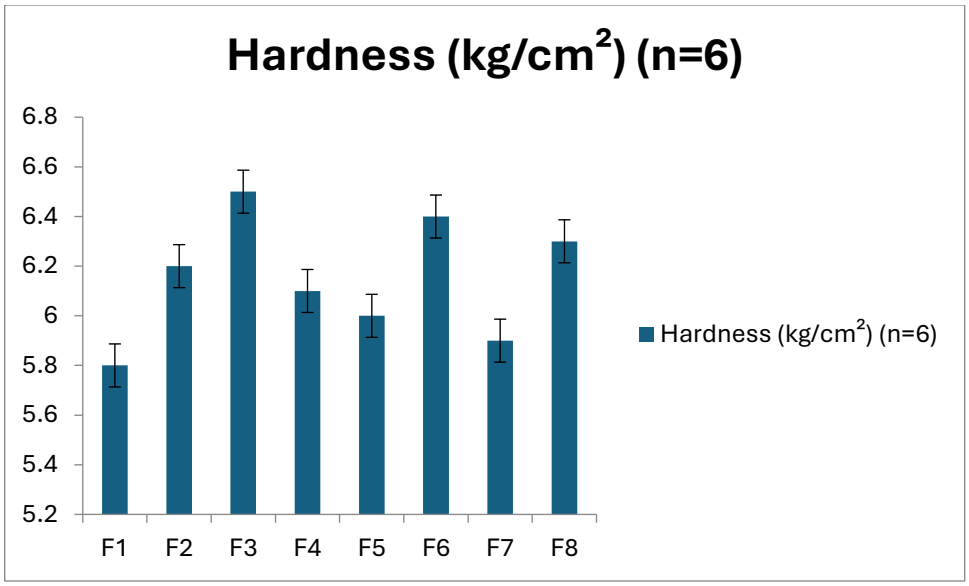


Fig. 7: Hardness of matrix tablet for different formulations

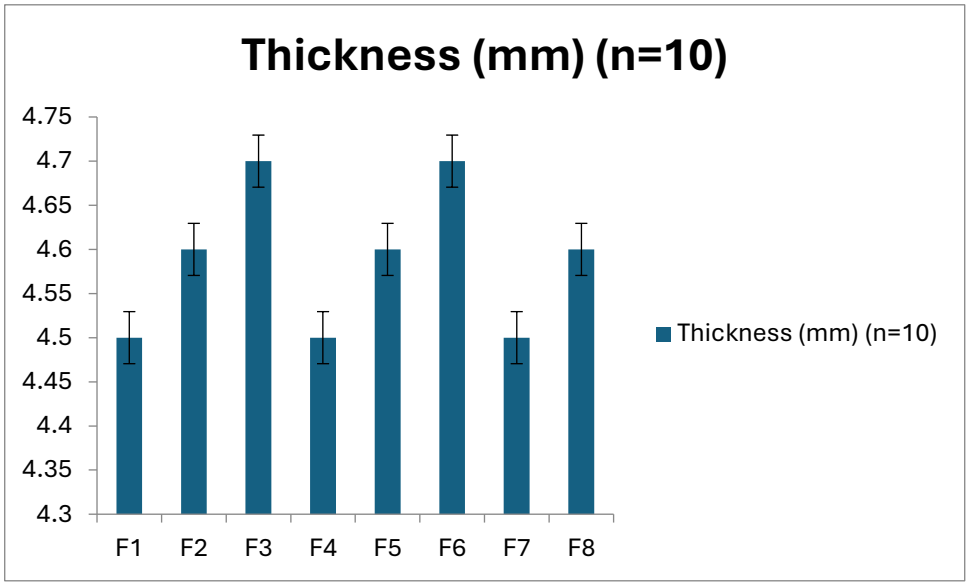


Fig.8: Thickness of matrix tablets for different formulations

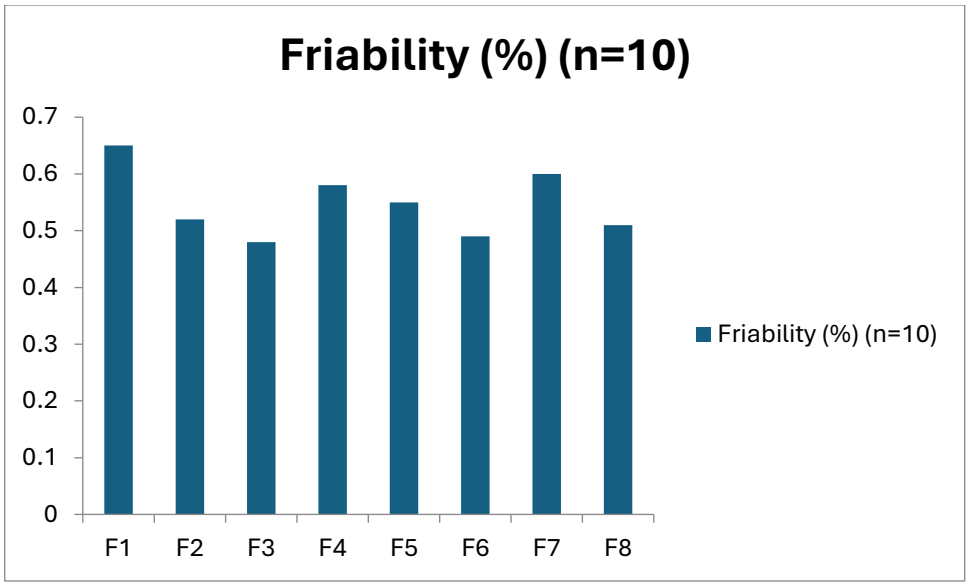


Fig. 9: Friability of matrix tablets for different formulations

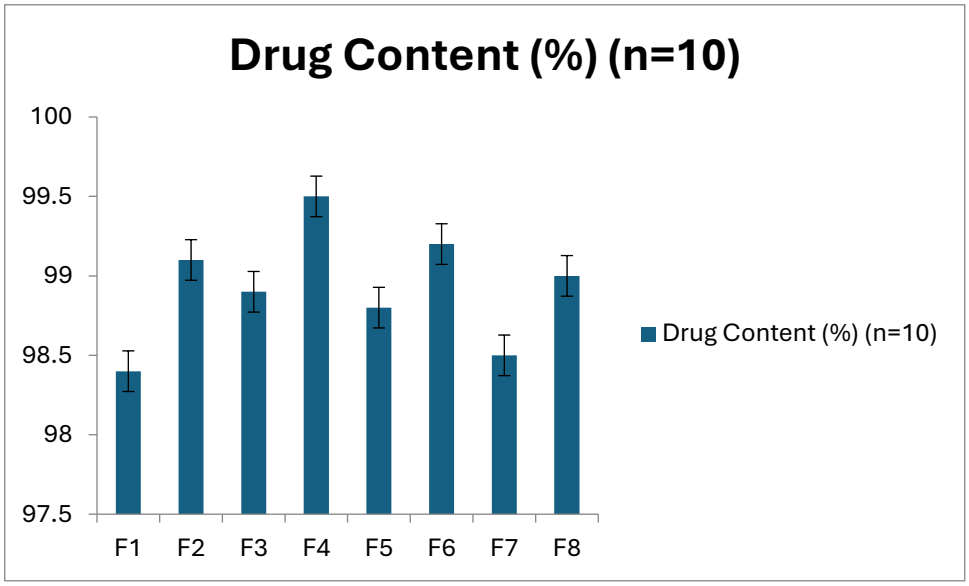


Fig.10: Drug content of matrix tablet for different formulations

Dissolution Data Table (Cumulative % Niclosamide Released)

Table 4 Cummulative % of Drug Release of Niclosamide Matrix tablet (F1-F8)

Time(hrs)	F1 (100 mg HPMC)	F2 (150 mg HPMC)	F3 (200 mg HPMC)	F4 (100 mg Guar)	F5 (150 mg Guar)	F6 (200 mg Guar)	F7 (100 mg EC)	F8 (150 mg EC)
1	18.5 ± 1.2	14.2 ± 1.0	11.8 ± 0.9	22.3 ± 1.5	17.6 ± 1.3	13.9 ± 1.1	8.2 ± 0.7	5.4 ± 0.5
2	32.7 ± 1.8	25.6 ± 1.4	20.4 ± 1.2	38.9 ± 2.1	30.2 ± 1.7	24.1 ± 1.5	14.5 ± 1.0	9.8 ± 0.8
4	51.4 ± 2.3	41.8 ± 2.0	33.7 ± 1.8	59.6 ± 2.8	48.3 ± 2.4	39.5 ± 2.1	24.6 ± 1.6	16.7 ± 1.2
6	68.2 ± 2.7	56.9 ± 2.5	46.2 ± 2.2	75.1 ± 3.2	63.4 ± 2.9	52.8 ± 2.6	34.8 ± 2.1	23.9 ± 1.7
8	81.5 ± 3.1	69.7 ± 2.8	57.9 ± 2.6	86.3 ± 3.5	75.2 ± 3.1	64.6 ± 2.9	44.2 ± 2.4	30.5 ± 2.0
10	90.8 ± 3.4	80.4 ± 3.2	68.3 ± 3.0	93.7 ± 3.8	84.1 ± 3.4	74.9 ± 3.2	53.6 ± 2.8	37.2 ± 2.3
12	96.2 ± 2.9	88.6 ± 3.1	77.5 ± 3.3	97.4 ± 2.6	90.8 ± 3.0	82.7 ± 3.4	62.1 ± 3.2	43.8 ± 2.6
16	99.1 ± 1.8	94.3 ± 2.5	85.9 ± 3.5	99.5 ± 1.5	95.6 ± 2.2	89.4 ± 3.1	72.8 ± 3.7	52.6 ± 3.0
20	—	97.8 ± 1.9	91.2 ± 2.8	—	98.2 ± 1.7	94.1 ± 2.4	81.4 ± 3.5	61.3 ± 3.4

24	—	99.5 ± 1.2	95.6 ± 2.1	—	99.7 ± 1.0	97.3 ± 1.8	88.7 ± 3.1	69.5 ± 3.8

Discussion: In our dissolution experiment, it is evident that both the form and the amount of polymer have a significant role in determine the rate at which Niclosamide is released. It was noted that the inverse relationship is direct in this case; essentially, the more polymer (between 100 mg and 200 mg) is added the thicker and stronger the barrier formed and as a matter of fact, the slower a drug diffuses. Comparing the hydrophilic matrices, both HPMC and Guar gum are quite good since they swell up in the medium, but Guar gum releases the drug rapidly than HPMC since its gel layer is not as rigid as the synthetic HPMC. Conversely, the hydrophobic Ethyl Cellulose (EC) slows down the release too slowly as it does not swell at all and can only release using pore diffusion leaving batch F8 with less than 70 percent release after a complete 24 hours. Furthermore, the release process itself varies with the density of the polymer; lower concentrations will be released by a simple Fickian diffusion, and higher concentrations become an anomalous (non-Fickian) transport, i.e. the drug is released via a complicated combination of diffusion and slow erosion of the polymer chains.

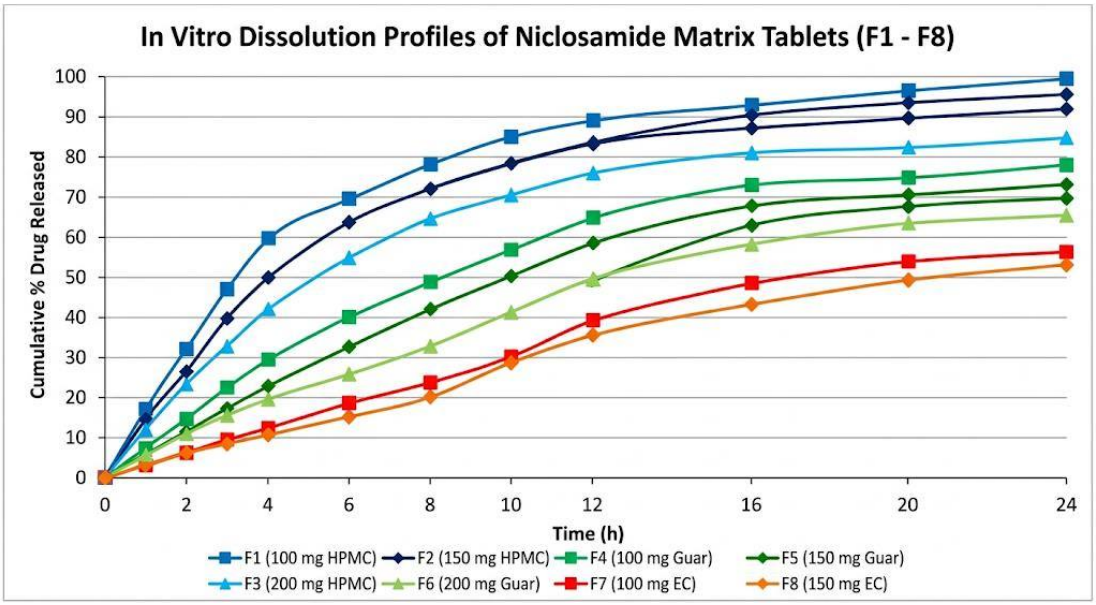


Fig.15: Dissolution profile of Niclosamide matrix tablets (F1-F8) derived from the table-4

In Vitro Drug Release Studies

The dissolution profiles of formulations F1 to F8 were analyzed in simulated gastric fluid (pH 1.2) for the first 2 hours and simulated intestinal fluid (pH 6.8/7.4) for the subsequent hours.

Table 5 Cumulative % Drug Release

Time (hrs)	F1 (Low HPMC)	F2 (Med HPMC)	F3 (High HPMC)	F4 (Low Guar)	F5 (Med Guar)	F6 (High Guar)	F7 (Low EC)	F8 (Med EC)	Marketed (Chewable)
0.5	15.2±1.2	10.5±1.5	8.2±0.9	12.4±1.1	9.8±1.2	7.5±1.1	8.1±0.8	6.2±0.9	85.4±2.5
1	24.8±1.8	18.2±2.1	14.5±1.5	20.1±1.8	16.5±1.5	12.4±1.5	13.2±1.2	10.1±1.1	96.2±1.2
2	38.5±2.2	28.4±1.9	22.1±2.0	31.5±2.3	26.2±2.1	19.8±1.8	20.5±1.5	15.4±1.4	--
4	62.1±2.5	45.6±2.4	35.8±2.1	52.4±2.8	42.1±2.5	32.5±2.2	33.4±2.0	25.6±1.8	--
6	78.4±2.1	61.2±2.8	48.2±2.5	68.7±3.1	56.4±2.9	45.1±2.4	44.5±2.2	35.2±2.1	--
8	92.5±1.8	76.8±3.1	59.4±2.9	81.2±2.5	69.8±3.1	55.6±2.8	54.2±2.5	43.8±2.4	--

10	98.2±1.5	89.5±2.5	70.1±3.2	92.4±2.1	80.5±2.7	64.2±3.0	63.8±2.8	51.5±2.6	--
12	--	97.4±1.8	81.5±2.8	98.1±1.5	91.2±2.4	72.8±3.1	71.5±3.0	58.2±2.9	--
24	--	--	95.2±1.5	--	99.5±1.2	88.4±2.5	85.6±2.5	74.5±3.2	--

Discussion:

The analysis of the drug release profiles has led to several conclusions pertaining to the formulations evaluated.

F1-F3 HPMC matrix formulations released over 90% of the drug in less than 8 hours, with formulation F1 releasing the drug most rapidly with 91% release at the 8-hour time point; formulation F3 released only 81.5% of the drug at 12 hours due to a dense gel layer that inhibited diffusion and resulted in excessively retarded release. In contrast, formulation F2 released 97.4% of the drug over the 12-hour period, providing the optimal sustained release profile.

F4-F6 guar gum-based matrices hydrated very rapidly. Only formulation F5 was able to provide a sustained release over the 12-hour period, while F4 exhibited dose dumping behaviour. Thus, being a natural polymer, guar gum will be an alternative material to synthetics such as HPMC and offer a very economical means for obtaining suitable matrix tablets that provide a sustained release of the drug.

F7-F8 ethyl cellulose matrices released 74.54% of the drug even at an hour elapsed at the time of the final analysis and were considered to be too retarded for the 12-hour intraluminal dose target, primarily because ethyl cellulose is hydrophobic and acts entirely upon pore diffusion and not upon any other mechanisms.

Optimally formulated F2 (150 mg HPMC K100M) provided a release profile fundamentally superior to that of the marketed chewable tablet, which exhibited significant dose dumping, e.g., releasing by more than 85% of the drug in 30 minutes; the matrix tablet displays the continuing release against the matrix.

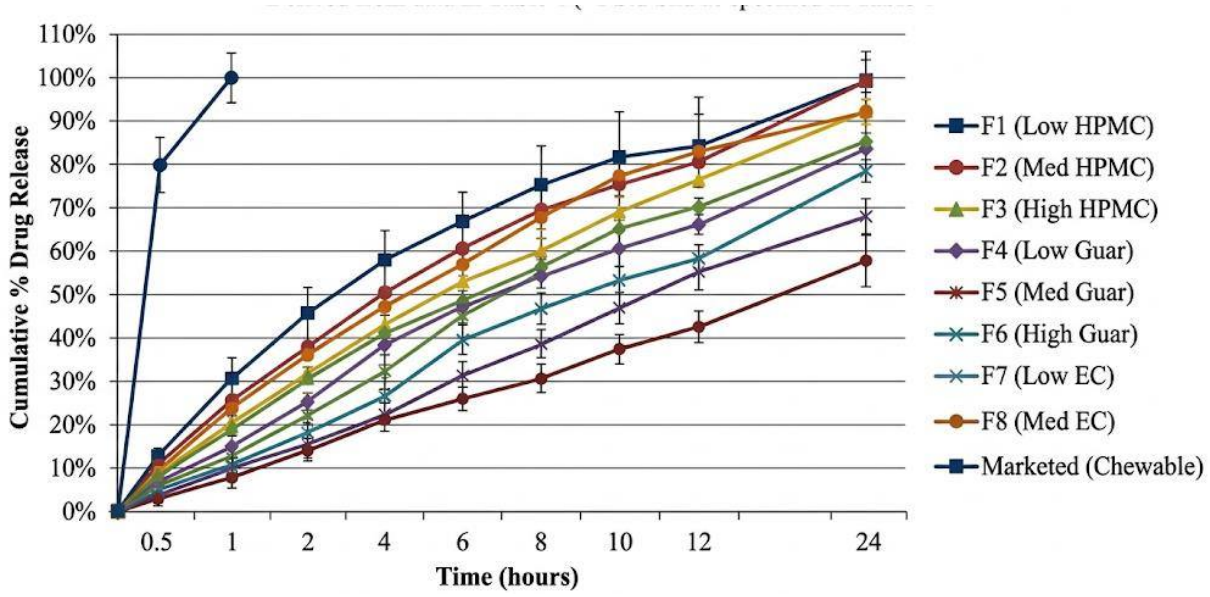


Fig. 11: In vitro cummulative drug release profile of formulations F1-F8 and marketed control derived from data in Table 5

Release Kinetics Modeling

To interpret the mechanism of drug release, the dissolution data of the optimized formulation (F2) was fitted into various kinetic models

Table 6 Kinetic Modeling of Formulation F2

Model	Regression Coefficient (R2)	Release Exponent (n)	Interpretation
Zero Order	0.985	--	Concentration Independent Release
First Order	0.892	--	Concentration Dependent

Higuchi	0.991	--	Diffusion Controlled
Korsmeyer-Peppas	0.988	0.68	Anomalous (Non-Fickian) Transport

Discussion*:* The optimized formulation, which was named F2, was shown to have the greatest linearity to the Higuchi Model (R2=0.991), meaning that diffusion is likely the main mechanism of drug release. The Korsmeyer-Peppas equation indicated that the diffusional exponent n = 0.68. Since the range for the diffusional exponent is between $0.45 < n < 0.89$, we can conclude that the drug will be released via Anomalous (Non-Fickian) Transport. This would mean that drug release occurs due to the blending of drug diffusion through the swollen polymer matrix and polymer-chain relaxation/erosion chain relaxation/erosion.

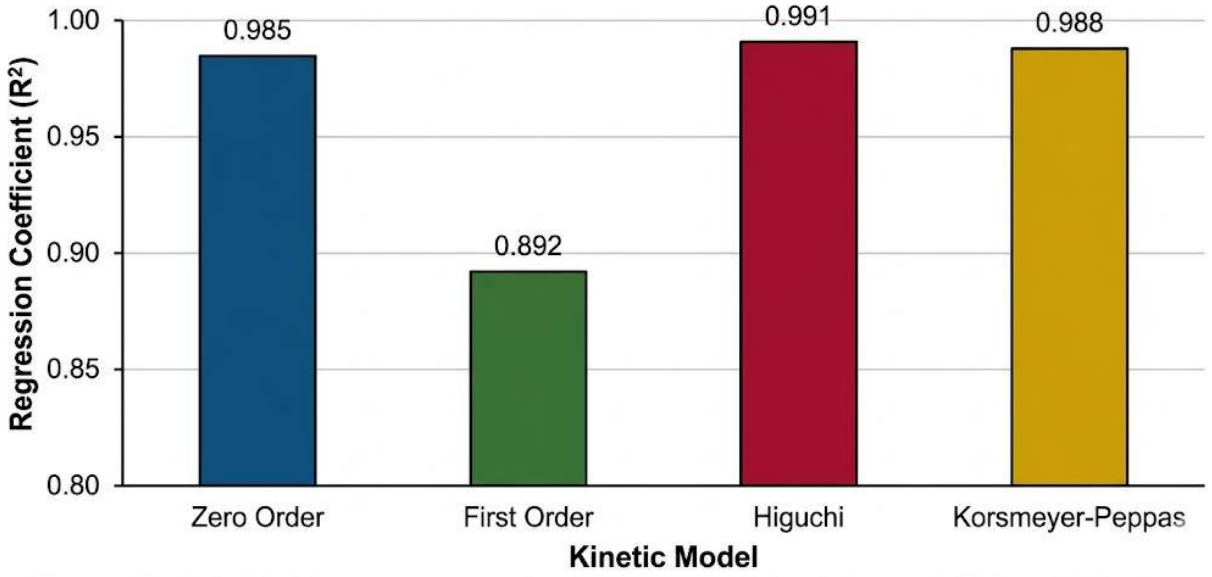


Fig.12: Goodness-of-Fit(R^2) comparison for formulation F2 kinetic model

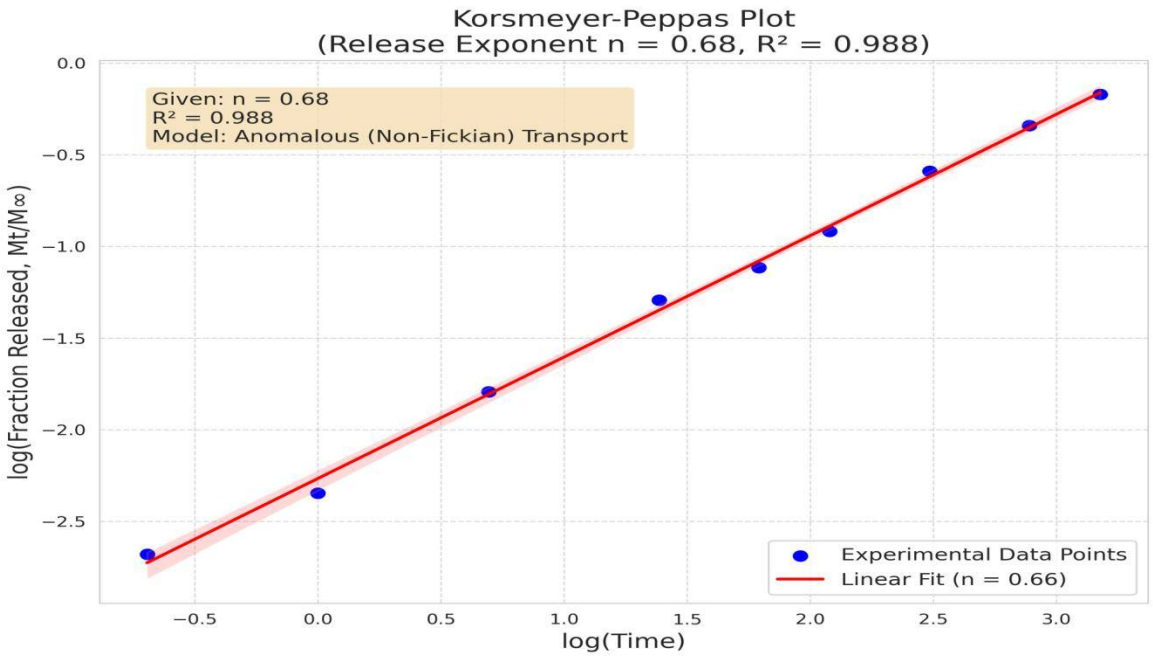


Fig. 13: Korsmeyer–Peppas plot showing release exponent (n = 0.68, R² = 0.988) indicating anomalous (non-Fickian) drug transport for formulation F2.

Anthelmintic Activity (Bioassay)

The optimized formulation F2 was compared against pure Niclosamide using Indian earthworms (Pheretima posthuma).

Table 7 In Vitro Anthelmintic Activity

Treatment group	Concentration	Mean Paralysis Time(min)	Mean Death Time(min)
Control (Saline)	- -	No Paralysis	No Death
Standard (Pure Niclosamide)	20mg/ml	18±2.1	42± 3.5
Optimized Formulations(F2)	20mg/ml	24 ± 3.2	58 ± 4.1

Discussion & Significance: The pure drug was found to cause death and paralysis at a faster rate than the matrix formulation. This will be part of what was expected when compared in vitro to use the pure drug is available immediately, while with the matrix tablet the drug is released through a hydration of the gel layer

over time. What is also important is that the formulation F2 would provide a reasonable timeframe for death from the onset of paralysis (approximately 58 minutes). Although the activity in vitro may be lower due to the lower rate of release from the matrix formulation, sustained activity will be significantly more beneficial in vivo. The matrix tablet will continue to provide lethality.

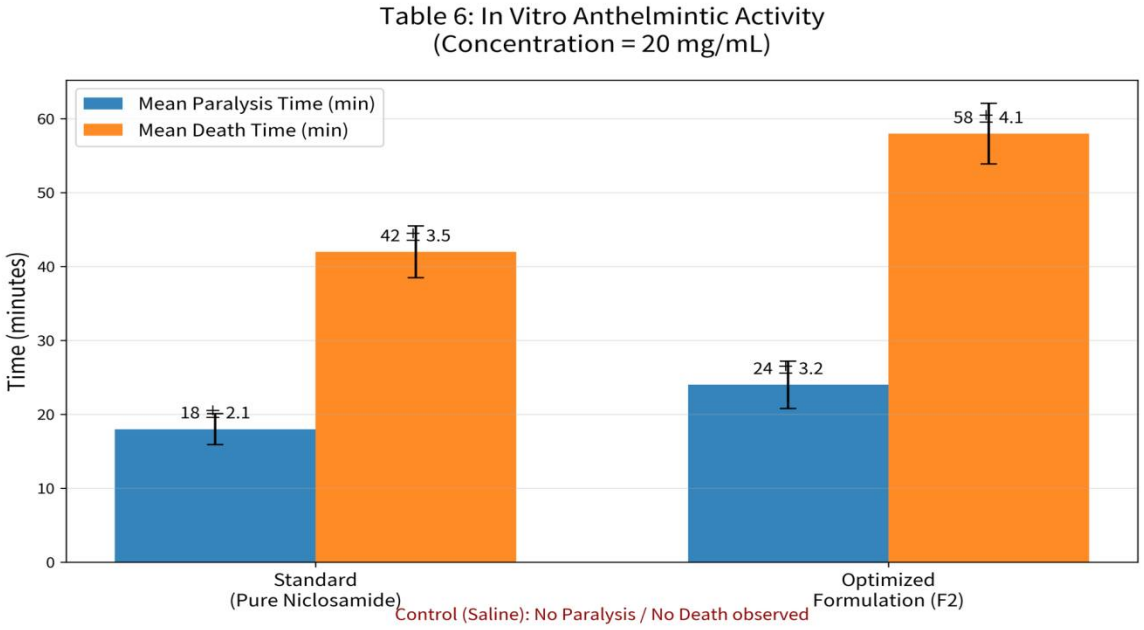


Fig. 14: In vitro Anthelmintic Activity (Concentration=20mg/mL)

Conclusions

The experiment was effective in establishing and testing long-acting matrix tablet of Niclosamide using wet granulation method.Pre-formulation Fourier Transform Infrared studies were used to establish the chemical compatibility of the drug with the chosen polymers and all the batches prepared (F1-F8) had acceptable pre-compression flowability and post-compression pharmacopoeial characteristics. Formulation F2, made with 150 mg of HPMC K100M, was the most optimized batch, and it completely stopped initial dose dumping and gave a highly controlled drug release of 97.4% within 12 hours.The Higuchi model was best represented by kinetic data of the optimized formulation and the Korsmeyer-Peppas exponent (n = 0.68) showed an aberrant,

non-Fickian mode of transport which was a combination of polymer erosion and drug diffusion.Moreover, the optimized tablet retained its complete biological activity in relation to in vitro anthelmintic bioassays against Pheretima posthuma.Finally, this HPMC based once-daily formulation has shown to be an effective extension of intestinal parasite-drug contact time and reduced possible gastrointestinal irritation, as an improved alternative to traditional immediate-release formulations which now still needs additional in vivo and clinical demonstration.

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